# Disease data - Documentation

Adapted from Narrative Report submitted by Epimetrics
Updated: 25 Jun 2024


**Tables** **1**
**Data Sources** **1**
**Data Request Processes and Data Sharing Agreements** **3**
DOH - Central 3
LGUs 3
**LGU Data Cleaning Notes** **4**
Muntinlupa 4
Palayan 5
Tacloban **5**
Zamboanga 5
Cagayan de Oro 5


## Tables

1. `disease_fhsis_totals`
   Total disease deaths (mortality) and cases (morbidity) reported by the Department of Health- Field Health Services Information System. Data is in city-monthly resolution.

2. `disease_pidsr_totals`
   Total disease cases (morbidity) reported in the Department of Health- Philippine Integrated Disease Surveillance and Response (DOH-PISDR) framework. Data is in city-weekly resolution.

3. `disease_psa_totals`
   Total disease-related deaths (mortality) recorded by the Philippine Statistical Authority (PSA) Vital Health Statistics division. Data is in city-weekly resolution.

4. `disease_lgu_disaggregated totals`
   Barangay-level age and sex-disaggregated disease deaths (mortality) and cases (morbidity) from reports submitted by the LGU to DOH-Field Health Services Information System (DOH-FHSIS) or Department of Health- Philippine Integrated Disease Surveillance and Response (DOH-PISDR). When `adm4_pcode` is not available, data is in a city level resolution.

# Data Sources

Sources of data for the health datasets `disease_pidsr_totals` and `disease_psa_totals` come from central/national government agencies, while `disease_fhsis_totals` and `disease_lgu_disaggregated totals` are sourced from LGUs.

Both PIDSR and FHSIS are surveillance systems, with PIDSR focused particularly on epidemiologic response (NEC-DOH, 2014), while FHSIS specializes on recording and reporting data that provides LGUS and the DOH with basis for the implementation of public health programs (DOH, 2018). In the usage of these two datasets, it is worth noting that PIDSR aggregate reports are incorporated into the FHSIS annual morbidity report (UP-NIH, 2011) and, as such, data from these two sources have been separated into two datasets to prevent double-counting in the linked dataset. The diseases available from each of these data sources are detailed in Table 1.

**Table 1.** Disease inclusions for each of the central/national data sources.

| | | **PIDSR** | **FHSIS** | | **PSA** |
|---|---|---|---|---|---|
| **ICD Code** | **Disease common name** | Cases | Cases | Deaths | Deaths |
| A09.0 | Acute Bloody Diarrhea | ☑ | ☑ | | |
| B15 | Hepatitis A | ☑ | | | ☑ |
| A00 | Cholera | ☑ | ☑ | | ☑ |
| A90-A91 | Dengue Fever | ☑ | ☑ | ☑ | ☑ |
| A27 | Leptospirosis | ☑ | ☑ | ☑ | ☑ |
| A82 | Rabies | ☑ | ☑ | | ☑ |
| A01 | Typhoid Fever | ☑ | ☑ | | ☑ |
| A08 | Rotaviral Enteritis | ☑ | | | |
| A92 | Chikunguya Viral Disease | ☑ | | | |
| E08-E14 | Diabetes | | ☑ | ☑ | ☑ |
| I11 | Hypertensive Heart Disease | | ☑ | ☑ | ☑ |
| I21-I22 | Ischemic Heart Disease | | ☑ | ☑ | ☑ |
| I63 | Ischemic Stroke | | | ☑ | |
| A09 | Gastroenteritis | | | | ☑ |
| J45 | Asthma | | ☑ | ☑ | ☑ |
| I20-I25 | Angina | | ☑ | | ☑ |
| T67 | Heat Stroke | | | | ☑ |

The spatiotemporal coverage of each of the sources also vary. Please refer to Table 2 for a quick summary of the included cities and dates.

| Source | Cities | Type | | Years Covered |
|---|---|---|---|---|
| | | Cases | Deaths | 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 |
| **PSA** | ALL | | ☑ | |
| **PIDSR-DOH** | ALL | ☑ | | |
| **PIDSR-LGU** Disaggregated | Tacloban City | ☑ | | |
| | Zamboanga City | ☑ | ☑ | |
| | Cagayan de Oro City | ☑ | | |
| **FHSIS-LGU** Totals only | Cagayan de Oro City | ☑ | | |
| | Muntinlupa City | | ☑ | |
| | Navotas City | ☑ | ☑ | |
| | Palayan City | | ☑ | |
| | Tacloban City | ☑ | | |
| | Zamboanga City | ☑ | | |
| **FHSIS-LGU** Disaggregated | Palayan City | ☑ | | |
| | Muntinlupa City | ☑ | | |
| | Cagayan de Oro City | ☑ | | |




**Table 2.** Spatiotemporal coverage of all health data sources.

# Data Request Processes and Data Sharing Agreements

## DOH - Central

Appropriate endorsements have been acquired from the DOH to request health data from both national/central government agencies and LGUs. DOH's endorsement for the project was acquired on 15 August 2024 and the Ethics Review Approval was obtained on 17 August 2024. Hospitalization datasets have also been requested from PhilHealth, but Data Usage Agreements (DUAs) have not been provided as of this writing.

## LGUs

In addition to the above central/national datasets, health data was also requested from the LGUs of the 12 cities involved in this project. DUAs were received from the following offices:

- Zamboanga CESU (City Epidemiology and Surveillance Unit)
- Tacloban CHO (City Health Office)
- Palayan CHO

- Muntinlupa CHO
- Navotas CHO
- Cagayan de Oro CHO

Data requests have also been submitted to other LGUs, but Epimetrics did not receive DUAs that would permit Project CCHAIN to open source their data. For the LGUs which we did not/have yet to receive data/DUA from, the decision/status is indicated in Table 3 as of May 2024.

**Table 3.** Data collection status from LGUs

| | |
|---|---|
| Mandaluyong | Health data from the previous administration is unavailable to the current administration, so the LGU has access only to data from 2020-2022 |
| Dagupan | Data requested but have yet to receive a response |
| Legazpi | Data is only available in monthly frequency and historical data is lost as physical copies have been damaged during various tropical storms. |
| Iloilo | Data received was limited. Data is not included because the DUA request is still pending with the LGU |
| Mandaue | DUA available, but data was not provided by LGU. |
| Davao | Data is available, but DUA request is still pending since March 2024. Hard copy of DUA is also with LGU, but no update was given to EpiMetrics since March. |

# LGU Data Cleaning Notes

Extensive QA was done to validate and error-correct the encoding for barangay level case and death totals

## Muntinlupa

Muntinlupa LGU data format was in scanned pdf. We used an OCR to detect tables.

- Muntinlupa 1
    - Page 2, Row 1 with month-week = 0; DISCARDED
- Muntinlupa 2
    - Page 5, Row 8 wrong ICD code A27 (Leptospirosis), everything else in file is B15 (Hepatitis A); Changed to B15 Hepatitis A
- Muntinlupa 3
    - Page 1, “NONE” in brgy; DISCARDED
    - Page 2, 0 in month; DISCARDED
    - Page 6, null brgy; DISCARDED
    - Page 11, blank brgy; DISCARDED

- Muntinlupa 4: Page 2, noted in pencil - “No rotavirus cases” - no action needed

## Palayan

- For the 2022 sheet, we discarded 56 rows with incomplete data, might likely be unfinished encoding.

## Tacloban

- Brgy 83-B (nonexistent) => matched to Brgy 83
- An age is encoded as ‘6A’, in hypertensive heart disease - DISCARDED because same as brgy, wrong encoding, ‘62B’, ‘60A’ changed to 60 bec of neighboring rows
- For cases with age in days => transformed to age in days and divided by 365 to get age in years

## Zamboanga

- Kaumpurnah Zone 1 is not in Zamboanga but in Isabela City, DISCARDED
- Kabasalan, Dipala is not in Zamboanga City but in Zamboanga Sibugay, DISCARDED

## Cagayan de Oro

- 100+ cases tagged to to barangay “POBLACION” but city has multiple brgys in its poblacion area - decided to include an extra fictitious brgy “POBLACION” with filler psgc “PH10430500x” to accommodate all unspecified poblacion brgys.
- Abbreviated street names dropped due to nonspecific information, or mapped to the following barangays
    - BUKI - dropped
    - TAGO - dropped
    - HAYE - likely Hayes St in POBLACION (PH10430500x)
    - COR, CORR - Corrales Ave, POBLACION (PH10430500x)
    - MONT - Pearlmont hotel? set as POBLACION (PH10430500x)
    - OS - likely Osmena St (PH10430500x)
    - COGON - Cogon Public Market (PH10430500x)
- The sheet “2018 Morbidity 2018 (clean).xlsx” - contains annual data, which is out of scope. This is discarded.